Long Data

Improving access to and use of the city's archival records

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Long data

Big data means many records, typically over a short span

Municipal records often run continuously for decades → *long data*

Long data can illustrate trends, establish precedents, enable comparisons, and uncover context

Long data also poses challenges around accessibility, standards, and comparability

Holding	Accessible?	Amount
Digital files with text	Yes	Small
Scanned pages	No	Large
Paper, film, audio	No	Unknown

Samuel Arbesman, "Stop Hyping Big Data and Start Paying Attention to 'Long Data'," *Wired*, January 29, 2013

Document understanding

Document understanding is a research field, and a set of software frameworks, that turns documents into data

- **Layout analysis** finds the headings, paragraphs, tables, and figures on a page, and the order to read them in
- **Optical character recognition (OCR)** reads printed or handwritten text from a picture of a page
- **Table recognition** rebuilds a table's rows, columns, and cells
- **Field extraction** pulls named items like dates or totals out of forms

AI models are rapidly improving at converting complex PDFs into accessible formats: text, Markdown, and HTML



From `https://www.datalab.to`

Data liberation and rescue

Data liberation moves information that is already public out of the formats that bury it (scans, PDFs, portals) and into open and accessible data formats

Data rescue is a grassroots, interdisciplinary movement to preserve public datasets from decay and censorship

- **2014.** The Sunlight Foundation runs a PDF Liberation Hackathon to build shared tools for pulling tables out of government PDFs
- **2017 on.** The Public Utility Data Liberation project turns federal energy filings into an open and documented database
- **2020.** The U.S. Supreme Court rules in *Georgia v. Public.Resource.Org* that the state's annotated code cannot be copyrighted
- **2025 on.** Researchers improvise technical capacity to archive data before it is deleted by the Trump administration

Tom Lee, "PDF Liberation: Why It Matters and How You Can Help," Sunlight Foundation, January 24, 2014; *Georgia v. Public.Resource.Org, Inc.*, 590 U.S. 255 (2020)

Data liberation for the City of Boulder

- **Hotline answers.** Helping staff research answers to questions from council and other departments
- **Budget history.** Converting PDF tables into accessible data formats
- **The law on a date.** Which version of an ordinance was in force when a permit was issued
- **Records requests.** A document made searchable once is ready for every later request
- **Accessibility.** Every scanned page gains text that a screen reader can speak
- **Research and reporting.** Residents, reporters, and researchers work from the same corpus as staff

Two data liberation sprints

I have already piloted two data liberation sprints using local government archives

City budget books, 2002–2027

City budgets

34 PDFs and 10,843 pages from city records

Six volumes needed OCR

450 values traced to source documents and reconciled

CDE enrollment yearbooks, 1986–2025

Pupil membership

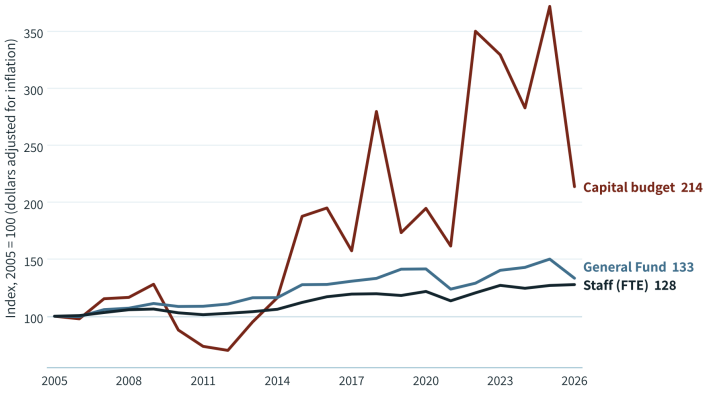
Scanned tables of enrollment by grade and school district

OCR and table extraction using Datalab

1,269 values extracted and audited for the Boulder Valley and St. Vrain Valley school districts

Agent pipelines identify page ranges, orchestrate OCR, extract tables, audit margins, record gaps, and compare across volumes

Twenty-two years of the city's budget



Indexed to 2005 = 100. Adopted budgets. General Fund: total uses. Capital: capital budget, all funds. Staff: FTE. Dollars in constant 2005 dollars (Denver–Aurora–Lakewood CPI-U). City of Boulder budget books and annual financial reports, via the budget-history dataset

Teaching with the city's records portal

Week 9 of my Web Data Science course uses the Finance Department's year-end revenue reports: `https://cuinfoscience.github.io/Web-Data-Science-Book/ch-09-pdfs.html`

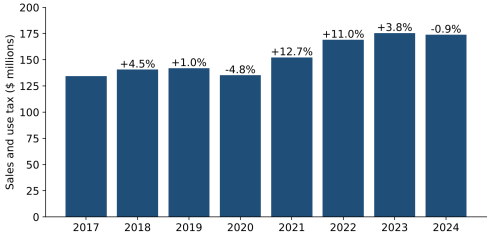
Students access the city's Laserfiche archive, download report PDFs, find and extract tables, and check the arithmetic against reported margins and prior years

Table 1 - Sales and Use Tax

All Funds

Sources - Related to Current Economic Activity	YTD through December			
	2017	2018	$ Change	% Change
Sales Tax	$104,144,137	$106,614,788	$2,470,651	2.37%
Rec MJ Additional 3.5%*	1,324,935	1,775,066	450,131	33.97%
Business Use Tax	14,216,145	13,070,617	(1,145,528)	(8.06%)
Construction Use Tax	9,897,738	13,018,049	3,120,311	31.53%
Motor Vehicle Use Tax	4,496,393	4,853,318	356,925	7.94%
Total Sales & Use Tax based on current economic activity	**$134,079,348**	**$139,331,838**	**$5,252,490**	**3.92%**
Sources - Related to Prior Periods				
Sales Tax Audit Revenue	189,929	211,748	21,819	11.49%
Business Use Tax Audit Revenue	269,615	1,105,124	835,509	309.89%
Total Sales and Use Tax	**$134,538,892**	**$140,648,710**	**$6,109,818**	**4.54%**

* Additional sales tax on Recreational Marijuana is dedicated to the general fund



Council hotline

Council members and staff exchange questions and answers via the council hotline

797 messages since March 2023: 209 questions from council and 144 answers from staff

Many of those answers point to budgets, presentations, ordinances, and reports already in the city's archives

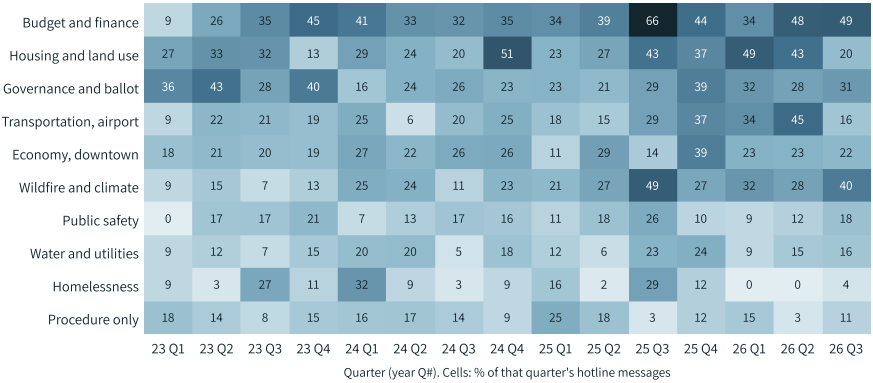
Question

How many of council's questions could be answered using the city's own records?

City of Boulder open data, "Emails to the City Council Hotline," March 2, 2023 to October 15, 2026

Analyzing hotline: questions from council

Classify each message's subject and body by keywords into nine policy themes



Code, keyword lists, and labels: `code/long-data-analysis.ipynb`, `code/hotline-labels-all.csv`. Messages through October 15, 2026

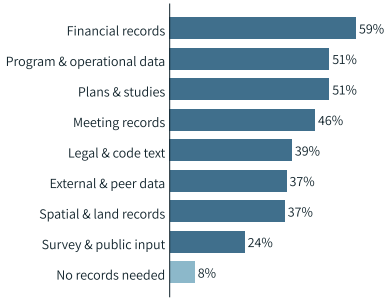
Analyzing hotline: answers from staff

Classify each staff answer's subject and body by keywords into eight kinds of records it draws on

Of 144 answers, the most common: plans and studies (47%), financial records (43%), program data (39%), legal and code text (38%)

56% draw on three or more kinds of records

Council's questions mention financial records more often (70%) than staff answers do (43%)



It goes without saying

An agent interface built on a document understanding pipeline should never replace staff expertise, experience, and oversight

What to liberate first

How often does council need the record? How far back does it run? How much personal information does it hold?

Start now

- Budgets, revenue, audits
- Council packets
- Staff memos
- Ordinances, resolutions
- Code versions
- Reports and plans

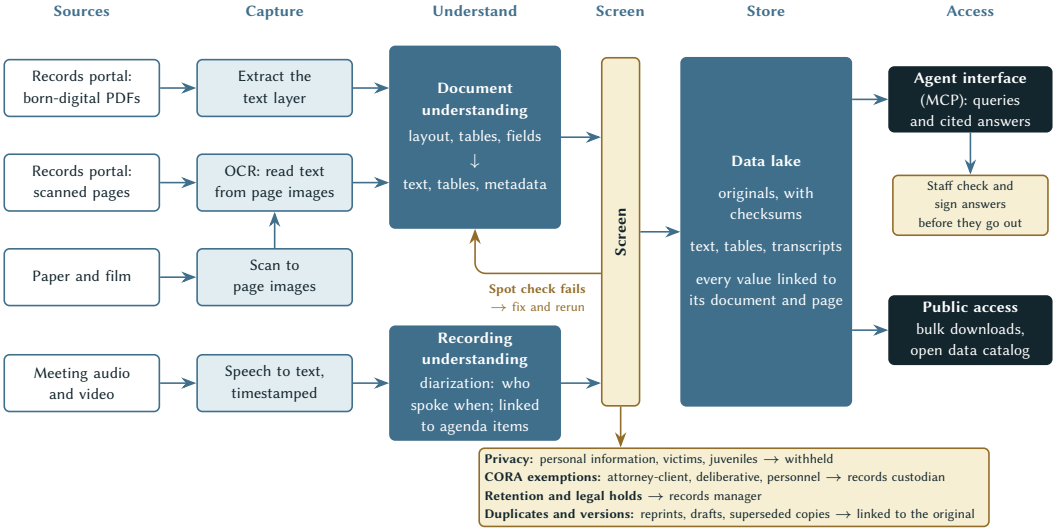
Next

- Minutes and recordings
- Boards and commissions
- Program year-end reports

Only with screening

- Permits and land records
- Contracts
- Emails

A proposed pipeline



Accessibility and CORA/FOIA

The ADA requires the city's web content, including PDFs, to meet accessibility guidelines (WCAG 2.1) by April 2027

Scanned documents without text layers and figures without alt text are endemic throughout municipal archives

Documents needed to get a service must be readable; archived scans are exempt

Fulfilling CORA requests can be expensive and complex → data liberation and rescue could save costs

Back of the envelope: OCR for all 10,843 budget-book pages at 0.75¢ a page is about $81, roughly two hours of research billed at CORA's $41.37 hourly cap



28 C.F.R. §§35.200–35.201, as amended by 91 Fed. Reg. 20902 (April 20, 2026); C.R.S. §24-34-802 (HB21-1110, HB24-1454); CORA, C.R.S. §24-72-201 et seq.

Community values

The city's Sustainability, Equity and Resilience (SER) Framework commits the city to a government that:

- "Models stewardship, resilience, and sustainability of the city's financial, human, **information**, and physical assets"
- "Supports strategic decision-making with … timely and **reliable data and analysis**"
- "Enhances and facilitates **transparency**, accuracy, efficiency, effectiveness"
- "Disaggregates data… to ensure programs, policies, and financial decisions do not have a disparate impact"

Uneven digitization and accessibility limit the ability of elected officials, staff, researchers, and journalists to study disparate impacts



What happens next

Profound improvements in performance per dollar using document understanding to support digitization, accessibility, transparency, and efficiency

Identify materials commonly requested by council and the public

Convert and document them in accessible formats through the existing Open Data team and infrastructure

Decisions about what to digitize, how to prioritize, where to archive, and who to open tooling to are unavoidably political

Protect staff time for substantive analysis and expert interpretation rather than menial information retrieval and cleaning

Implement SER Framework values through open standards and products, not expensive and fragile vendor solutions



Why I might be wrong

Major pivot in data culture requiring procurement, training, and maintenance

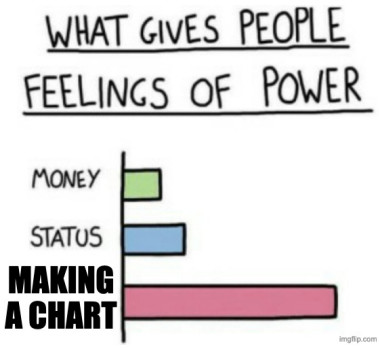
Demand for agentic interfaces into municipal archives may be low — or introduce novel political, legal, and operational risks

Early mover risks with dramatic changes in architectures, standards, and practices for AI, agents, MCP servers, and data lakes

Staff may already find most answers with the search tools they have, and know where to look

The records may not hold the answer: in 40 staff answers I coded by hand, half needed judgment or new analysis

Models write plausible wrong numbers, and one wrong figure in an answer to council costs more trust than the time it saved



Other risks

Documents in archives fall into at least three categories: **already data**, **could become data**, or **shouldn't become data**

Surveillance video, customer records, code complaints, and records about victims and juveniles should *not* become data

What gets scanned gets cited → paper archives omitted from trends and comparisons

OCR does worst on old, handwritten, tabular, and non-English pages

Use the SER Framework to guide digitization priorities



Key takeaways

1. The city holds long data of immense value locked away in PDFs and paper
2. Digitizing is getting better and cheaper, while the cost of verifying records stays roughly constant
3. The hotline analysis shows more than half of staff answers draw on three or more kinds of records
4. Accessibility law, CORA, and the city's own framework already call for this work

Thank you

Source code and data for this deck:
`https://github.com/CUPIDS-Lab/city-of-boulder/tree/main/2026-10-21-ett`

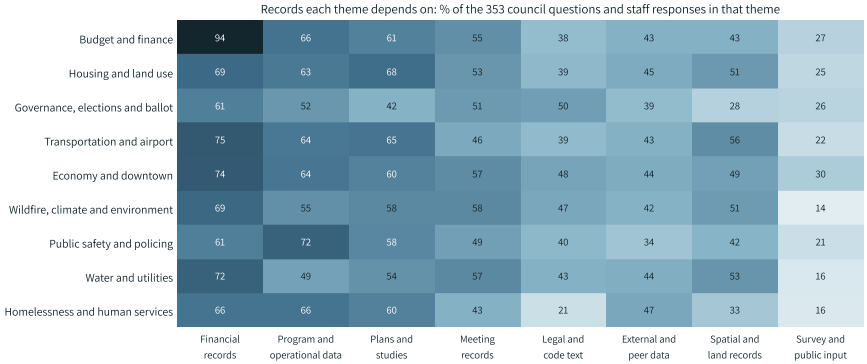
Claude Opus 5.5 was used for the hotline data analysis

Questions? brian.keegan@colorado.edu or me@brianckeegan.com

Appendix: record priorities in detail

Record	Serves	What we know of its state	Tier
Budget books	Financial questions; 26-year series	2002–2027 in the records portal; 1,651 pages lack text	Start
Revenue reports	Financial questions	In the records portal, 2018–2024; also sent as hotline attachments, which are not in the dataset	Start
Council packets and memos	Meeting records; plans; the answers themselves	Recent years in the agenda system; depth unknown	Start
Ordinances, resolutions, code versions	Legal text; which law applied when	Current code hosted outside the city; history in the records portal	Start
Minutes and recordings	Deliberation and intent	Unknown; needs inventory	Next
Plans and consultant studies	Plans and studies	Comprehensive plan: 15 editions converted, 1977–2026	Next
Board and commission appointments	Who served on what, when	Announced on the hotline; rosters unknown	Next
Program year-end reports	Operational data	Some sent as hotline attachments	Next
Permits and land records	Where things are	Department systems; custody split	Screen
Hotline and public comment	What council asks; deliberation	904 records, March 2023 to October 15, 2026; email addresses and phone numbers coded	Screen

Appendix: which records each theme depends on



Financial records come up in more than 60% of questions and answers in every theme. Most answers need several kinds of record at once